The lecture hall quadratic assignment problem

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1 Introduction and background

Universities and colleages often determine the courses timetable well before they assign lectures to specific halls. Once the timetable is fixed and student registration is completed, the institution knows both the size of each lecture and, to some extent, the overlap in enrollment between consecutive lectures. At that stage, the hall-assignment decision can have a substantial effect on student mobility across campus. Assigning consecutive lectures attended by many of the same students to distant halls may generate unnecessary walking, even when every lecture is assigned to a feasible room. In practice, long walks between halls may also cause students to arrive late at their next class, thereby disrupting the beginning of lecture. It may also be particularlly challanging to students with mobility disablities.

This observation motivates a lecture-hall assignment problem that is separate from the timetabling itself. In the lecture-hall assignment problem, the timetable, including the scheduled time interval of each lecture, is given. The remaining task is to assign lectures to halls in a way that respects room capacities and temporal compatibility, while also reducing the walking burden imposed on students as they move between consecutive classes. The proposed model is related to three main streams of literature: the quadratic assignment problem (QAP), classroom assignment and interval scheduling, and university course timetabling. The closest conceptual link to the QAP is the presence of pairwise interaction costs: assigning two lectures to two halls creates a cost that depends jointly on the distance between the halls and on the number of students who attend both lectures. This is analogous to the flow–distance structure that defines the classical QAP introduced by Koopmans and Beckmann (1957) and surveyed extensively by Burkard et al. (1998) and Loiola et al. (2007). At the same time, the present problem differs from the standard QAP because feasibility is driven by temporal overlap, room capacities, and the subset of lecture pairs that are consecutive on the same day.

A second relevant stream concerns classroom assignment and interval-based room allocation. In this family of problems, activities with fixed or partially fixed time intervals must be assigned to rooms while respecting incompatibilities caused by overlap in time. Carter and Tovey (1992) study the classroom assignment problem from a complexity perspective and explicitly connect it to interval scheduling. Their framework is closely related to the feasibility component of the present model, where lectures cannot share a hall if their time intervals overlap. However, the objective considered

here is richer because it is not limited to feasibility or room utilization; instead, it introduces interaction costs across consecutive lectures through student flows and hall-to-hall distances.

A third stream is the university timetabling literature, which studies the joint allocation of courses to times and rooms under institutional constraints. General surveys by Schaerf (1999) and Burke and Petrovic (2002) describe a broad range of formulations, including problems in which room capacities, conflicts, and soft preferences play a central role, while the more recent review of Abdipoor et al. (2023) summarizes modern meta-heuristic and hybrid approaches for university course timetabling. Within this literature, classroom assignment is often treated either as a subproblem of course timetabling or as a second-stage problem after times have been fixed. For example, Torres-Ovalle et al. (2014) propose a two-phase approach in which course scheduling is followed by classroom assignment, and Rappos et al. (2022) develop a mixed-integer programming model that jointly considers student allocation, room assignment, and time-period assignment. More recently, Davison et al. (2025) extend university timetabling models to hybrid teaching environments, showing how current formulations continue to expand in response to operational changes in higher education. This decomposition is natural in many university settings and is also compatible with the present formulation, which assumes that the scheduled time interval of each lecture is already known.

Taken together, these three literatures suggest that the lecture hall assignment problem studied here combines structural ingredients that are usually treated separately. From classroom assignment and interval scheduling it inherits the temporal packing constraints on rooms; from timetabling it inherits the academic context, capacity restrictions, and interpretation of lectures as scheduled activities; and from the QAP it inherits a pairwise objective that couples assignment decisions across lectures. A particularly close recent connection is the room-allocation study of Antunes-Batista et al. (2026), which also considers a fixed-timetable room-assignment subproblem and explicitly accounts for student movement between consecutive classes. Their model is multiobjective, simultaneously penalizing student relocation between buildings, room-feature mismatches, capacity waste, and room changes, and it is solved primarily by customized multiobjective evolutionary algorithms with local search, benchmarked against a penalized ILP model. The present framework can likewise accommodate mandatory room features through the compatibility sets and additional room-quality criteria through linear assignment penalties. Its distinguishing feature, however, is that student movement enters as a QAP-style quadratic term induced by shared enrollments and inter-hall walking distances, with capacity and overlap kept as hard feasibility constraints inside a single exact optimization model. This distinction yields a different mathematical structure and motivates exact mathematical-programming and constraint-programming approaches rather than a multiobjective metaheuristic framework.

Research gap and contribution. Although recent work in university timetabling and room allocation accounts for room capacities, time conflicts, and in some cases student-relocation objectives, these decisions are usually embedded in broader timetabling models or treated as one component of a multiobjective room-allocation problem; see, for example, Rappos et al. (2022), Davison et al. (2025), and (Antunes-Batista et al., 2026). In contrast, we study a fixed-time lecture-hall assignment problem in which the timetable is given and the dominant structural component is a QAP-style quadratic walking term induced by shared enrollments and inter-hall distances. The framework keeps capacity and overlap as hard constraints, and it can also incorporate mandatory

room features through the compatibility sets together with additional linear assignment penalties when desired. This setting is natural in institutions where timetabling decisions are made before registration is finalized and actual course enrollments are then observed. Isolating the hall-assignment layer yields a distinct exact optimization problem with exploitable structure. We analyze its computational complexity, derive mathematical programming and constraint programming formulations together with valid inequalities and safe preprocessing techniques, and show in the numerical study that these enhancements substantially improve computational performance relative to the baseline nonlinear formulation while enabling the solution of large and realistic instances to optimality or near-optimality.

2 Problem statement and complexity

We now formalize the lecture-hall assignment problem and state its complexity properties.

We consider this problem through the following sets:

Halls (H) Each hall is defined by its capacity and location.

Lectures (L) Each lecture is defined by its scheduled time interval and its number of students.

Immediate-successor relation (A) A relation $A \subseteq L \times L$, where $(l_1, l_2) \in A$ if lecture l_2 is scheduled immediately after lecture l_1 for at least one group of students.

The walking distance between each pair of halls $h_1, h_2 \in H$ is given by the distance matrix d_{h_1,h_2}. In addition, for each pair of lectures $(l_1, l_2) \in A$, we are given the number of students enrolled in both classes, denoted by c_{l_1,l_2}. This number may be a simple head count or a score that gives higher weight to students with disabilities, if the institution has this information at its disposal.

The goal is to assign each lecture to a hall with sufficient capacity, such that no two lectures assigned to the same hall overlap in time. The objective is to minimize the total walking distance walked by students between consecutive classes.

This problem shares features with both the quadratic assignment problem and interval scheduling, and in a sense it combines elements of both.

A natural decision version of the lecture-hall assignment problem asks whether, for a given bound B, there exists a feasible hall assignment whose total walking cost is at most B. This decision problem belongs to NP, since a proposed assignment can be verified in polynomial time by checking compatibility, non-overlap, and the resulting objective value. The next proposition states the exact and approximation hardness of the problem.

Proposition 1. *The decision version of the lecture-hall assignment problem is NP-complete, and the corresponding optimization problem is NP-hard. Moreover, for every $\rho \geq 1$, if there exists a polynomial-time ρ-approximation algorithm for the lecture-hall assignment problem, then there exists a polynomial-time ρ-approximation algorithm for the classical quadratic assignment problem*

(QAP). Consequently, every approximation lower bound for QAP carries over to the lecture-hall assignment problem. All of these statements remain true even for a restricted single-day case with two consecutive time blocks, identical hall capacities, and full lecture–hall compatibility.

Proof. We use a reduction from the classical QAP, which is NP-hard and in approximable with a constant factor, see Sahni and Gonzalez (1976); Burkard et al. (1998); Loiola et al. (2007). Consider a QAP instance with facilities $F = \{1, \ldots, n\}$, locations $R = \{1, \ldots, n\}$, nonnegative flow matrix (f_{ij}), and nonnegative distance matrix (d_{ab}). First normalize the distances by setting

$$\hat{d}_{aa} = 0 \quad \forall a \in R, \qquad \hat{d}_{ab} = d_{ab} + 1 \quad \forall a \neq b.$$

For every permutation π of the locations,

$$\sum_{i=1}^{n} \sum_{j=1}^{n} f_{ij} \hat{d}_{\pi(i),\pi(j)} = \sum_{i=1}^{n} \sum_{j=1}^{n} f_{ij} d_{\pi(i),\pi(j)} + \sum_{i \neq j} f_{ij},$$

because $\pi(i) \neq \pi(j)$ whenever $i \neq j$. Thus the normalization adds the same constant to every feasible QAP solution, so it preserves both exact comparisons and approximation ratios.

Now construct a lecture-hall instance as follows. Create n halls $h_1, \ldots, h_n$, all with identical capacity and all compatible with every lecture. Create two sets of lectures,

$$U = \{u_1, \ldots, u_n\} \qquad \text{and} \qquad V = \{v_1, \ldots, v_n\},$$

where all lectures in U occupy one common time block and all lectures in V occupy the immediately following common time block. Every lecture in U therefore overlaps every other lecture in U, and similarly for V. Since there are exactly n halls, every feasible lecture-hall assignment induces two permutations of the halls: one permutation π for the lectures in U and one permutation σ for the lectures in V. For every $i, j \in \{1, \ldots, n\}$, include the pair (u_i, v_j) in the successor set and set

$$d_{h_a,h_b} = \hat{d}_{ab} \qquad \forall a, b \in \{1, \ldots, n\}.$$

Let M be a positive penalty parameter, to be specified below, and define

$$c_{u_i,v_j} = \begin{cases} f_{ij} + M, & \text{if } i = j, \\ f_{ij}, & \text{if } i \neq j. \end{cases}$$

Under a feasible lecture-hall assignment corresponding to permutations (π, σ), the objective value is

$$\sum_{i=1}^{n} \sum_{j=1}^{n} f_{ij} \hat{d}_{\pi(i),\sigma(j)} + M \sum_{i=1}^{n} \hat{d}_{\pi(i),\sigma(i)}.$$

If $\pi \neq \sigma$, then there exists some i such that $\pi(i) \neq \sigma(i)$, and therefore $\hat{d}_{\pi(i),\sigma(i)} \geq 1$. Hence every such assignment has value at least M. On the other hand, if $\pi = \sigma$, then the penalty term vanishes and the lecture-hall objective becomes

$$\sum_{i=1}^{n} \sum_{j=1}^{n} f_{ij} \hat{d}_{\pi(i),\pi(j)},$$

which is exactly the normalized QAP objective.

For the exact reduction, start from the QAP decision problem with threshold K and choose

$$M > \hat{K} := K + \sum_{i \neq j} f_{ij}.$$

Then any lecture-hall solution of value at most $\hat{K}$ must satisfy $\pi = \sigma$, and once $\pi = \sigma$ its value is exactly the normalized QAP value of the same permutation. Therefore the original QAP instance is a yes-instance if and only if the constructed lecture-hall instance admits a feasible assignment of cost at most $\hat{K}$. The reduction is polynomial, so the decision version of the lecture-hall assignment problem is NP-hard. Since it belongs to NP, it is NP-complete. Consequently, the optimization problem is NP-hard.

For the approximation statement, fix $\rho \geq 1$ and let

$$F_{\text{tot}} = \sum_{i=1}^{n} \sum_{j=1}^{n} f_{ij}, \qquad D_{\max} = \max_{a,b \in R} \hat{d}_{ab}, \qquad U_{\max} = F_{\text{tot}} D_{\max}.$$

Every feasible normalized QAP solution has value at most $U_{\max}$, and therefore the optimal value OPT_{QAP} satisfies $\text{OPT}_{\text{QAP}} \leq U_{\max}$. Now choose

$$M = \rho U_{\max} + 1.$$

Then every lecture-hall solution with $\pi \neq \sigma$ has value at least

$$M = \rho U_{\max} + 1 > \rho U_{\max} \geq \rho \, \text{OPT}_{\text{QAP}}.$$

But the optimal lecture-hall value is exactly OPT_{QAP}, attained by solutions with $\pi = \sigma$. Therefore any polynomial-time ρ-approximation algorithm for the lecture-hall assignment problem, applied to this constructed instance, must return a solution with $\pi = \sigma$, because every solution with $\pi \neq \sigma$ already exceeds ρ times the optimum. For such a returned solution, the lecture-hall objective is exactly the normalized QAP objective of the same permutation. Hence a polynomial-time ρ-approximation for the lecture-hall assignment problem yields a polynomial-time ρ-approximation for QAP.

Thus both exact hardness and approximation hardness follow from the same reduction family. □

3 Models and algorithms

This section presents three complementary exact modeling views of the lecture-hall assignment problem. We begin with a natural nonlinear integer programming formulation that exposes the structure of the problem and supports several structural observations and preprocessing ideas. We then derive a compact mixed-integer programming reformulation with strengthening inequalities, and finally present a constraint-programming formulation that captures the assignment structure directly.

For notational convenience, define

$$A' = \{(l_1, l_2) \in A : c_{l_1,l_2} > 0\}.$$

3.1 Nonlinear Integer Programming Formulation

A natural nonlinear integer programming formulation is as follows. For each lecture $l \in L$ and each compatible hall $h \in H(l)$, let

$$x_{lh} = \begin{cases} 1 & \text{if lecture } l \text{ is assigned to hall } h, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

Let s_l denote the number of students attending lecture l, and let u_h denote the capacity of hall h. Define $H(l) \subseteq H$ as the set of halls that are compatible with lecture l. In the basic model studied here, compatibility is determined only by capacity, so

$$H(l) = \{h \in H : u_h \geq s_l\}.$$

In addition, let T be the set of elementary time units in the planning horizon, and for each $t \in T$ let $L(t) \subseteq L$ denote the set of lectures whose scheduled intervals include time unit t. The problem can then be written as

$$\min \quad \sum_{(l_1,l_2)\in A'} c_{l_1,l_2} \sum_{h_1\in H(l_1)} \sum_{h_2\in H(l_2)} d_{h_1,h_2}\, x_{l_1h_1} x_{l_2h_2} \tag{2}$$

$$\text{s.t.} \quad \sum_{h\in H(l)} x_{lh} = 1, \qquad \forall l \in L, \tag{3}$$

$$\sum_{l\in L(t)} x_{lh} \leq 1, \qquad \forall t \in T,\ \forall h \in H, \tag{4}$$

$$x_{lh} \in \{0,1\}, \qquad \forall l \in L,\ \forall h \in H(l). \tag{5}$$

The objective is quadratic because the cost of assigning a lecture to a hall depends on where the immediately succeeding lecture is assigned. The first set of constraints ensures that each lecture is assigned to exactly one compatible hall, and the second is a stronger non-overlap condition: for every hall and every time unit, at most one lecture occupying that time unit may be assigned to that hall.

An immediate structural observation is that the problem is separable by day. Let D denote the set of teaching days, and for each $d \in D$ let $L_d \subseteq L$ be the set of lectures scheduled on day d. Likewise, let $A_d \subseteq A$ be the set of successor pairs whose two lectures both belong to day d, and let $T_d \subseteq T$ be the set of time units associated with day d. Because temporal-overlap constraints involve only lectures that occur on the same day, and because an immediate-successor pair must also lie within a single day, both the objective and feasibility constraints decompose across days. Therefore, the weekly problem can be written as the sum of independent daily subproblems,

$$\min \sum_{d\in D} f_d(x^d), \tag{6}$$

where each f_d is the objective induced by the lectures and successor pairs of day d, and x^d denotes the hall-assignment decisions for that day. In particular, solving the overall weekly problem is equivalent to solving one independent lecture-hall assignment problem for each day and combining the resulting assignments. This observation is useful both computationally and conceptually. Therefore, in what follows one may interpret the model as applying either to the full week or, equivalently, to a single-day instance.

The interval structure also yields a tighter way to express temporal feasibility. Fix a day $d \in D$, and define the interval graph $G_d = (L_d, E_d)$ in which two lectures are adjacent if and only if their scheduled intervals overlap. Since the lecture intervals lie on a line, G_d is an interval graph. Let $\mathcal{C}_d$ denote the family of maximal cliques of G_d. Then the slot-indexed constraints (4) may be replaced by the clique constraints

$$\sum_{l \in C:\, h \in H(l)} x_{lh} \leq 1, \qquad \forall d \in D,\ \forall C \in \mathcal{C}_d,\ \forall h \in H. \tag{7}$$

This replacement is exact. For every time unit t, the set $L(t)$ of lectures active at time t forms a clique in the interval graph, and therefore is contained in a maximal clique. Conversely, by the Helly property of intervals, every clique of an interval graph has a common intersection point, so each maximal clique corresponds to a set of lectures that are simultaneously active at some time. Hence (7) is equivalent to the family of slot-indexed non-overlap constraints, but it is often smaller because many consecutive time units induce identical or nested active sets. The same observation applies to the MILP and CP formulations below: one may replace the time-indexed overlap constraints, or the corresponding `AllDifferent` constraints, by maximal-clique constraints of the interval graph.

A useful preprocessing step can shrink the compatibility sets $H(l)$ before solving any of the three formulations below. Fix a target lecture $l' \in L$ and consider the auxiliary problem

$$\begin{aligned}
U_{l'} = \max \quad & \sum_{h \in H(l')} u_h x_{l'h} \\
\text{s.t.} \quad & \sum_{h \in H(l)} x_{lh} = 1, && \forall l \in L, \\
& \sum_{l \in L(t)} x_{lh} \leq 1, && \forall t \in T,\ \forall h \in H, \\
& x_{lh} \in \{0,1\}, && \forall l \in L,\ \forall h \in H(l).
\end{aligned}$$

This model keeps only the hard feasibility constraints and maximizes the capacity assigned to the single lecture l'. If $U_{l'}$ denotes its optimal value, then every hall $h \in H(l')$ with $u_h > U_{l'}$ may be removed from $H(l')$.

Indeed, suppose that some globally feasible assignment uses lecture l' in a hall $\hat{h}$ with $u_{\hat{h}} > U_{l'}$. That same assignment satisfies the preprocessing constraints above and attains objective value at least $u_{\hat{h}}$, contradicting the definition of $U_{l'}$. Therefore the reduced set

$$\widehat{H}(l') = \{h \in H(l') : u_h \leq U_{l'}\}$$

preserves all feasible solutions of the original problem. Repeating this optimization for each lecture yields a valid preprocessing step for the MIQP, the compact MILP, and the CP model alike.

One may also use a cheaper lightweight version. Let

$$L'(l') = \{l'\} \cup \{l \in L : l \text{ overlaps } l' \text{ in time}\},$$

and solve the same maximization problem with $L'(l')$ replacing L and with the overlap constraints restricted accordingly. Let its optimal value be $\widetilde{U}_{l'}$. Since the lightweight problem removes lectures and therefore relaxes the feasible region, one has

$$\widetilde{U}_{l'} \geq U_{l'}.$$

Consequently, deleting from $H(l')$ only the halls with capacity larger than $\widetilde{U}_{l'}$ is still safe: it may leave more halls than the full preprocessing, but it cannot remove a hall that is needed by a globally feasible solution.

3.2 Mixed-Integer Programming Reformulation

A compact linearization can be obtained by introducing, for each successor pair $(l_1, l_2) \in A$ such that $c_{l_1,l_2} > 0$, one nonnegative continuous variable $z_{l_1 l_2}$ representing the realized hall-to-hall distance induced by that pair. The problem can then be reformulated as the mixed-integer linear program

$$\begin{aligned}
\min \quad & \sum_{(l_1,l_2)\in A'} c_{l_1,l_2} z_{l_1 l_2} && && (8)\\
\text{s.t.} \quad & \text{Constraints (3) and (4)} \\
& z_{l_1 l_2} \geq d_{h_1,h_2}\big(x_{l_1 h_1} + x_{l_2 h_2} - 1\big), && \forall (l_1,l_2) \in A', h_1 \in H(l_1), h_2 \in H(l_2), && (9)\\
& x_{lh} \in \{0,1\}, && \forall l \in L,\ \forall h \in H(l), && (10)\\
& z_{l_1 l_2} \geq 0, && \forall (l_1,l_2) \in A'. && (11)
\end{aligned}$$

This reformulation is exact. Since each lecture is assigned to exactly one hall, for every pair (l_1, l_2) there is exactly one compatible hall pair (h_1, h_2) for which $x_{l_1 h_1} = x_{l_2 h_2} = 1$; for that pair, constraint (9) yields

$$z_{l_1 l_2} \geq d_{h_1,h_2}.$$

For every other hall pair, the right-hand side of (9) is nonpositive. Therefore, because the objective minimizes $\sum c_{l_1,l_2} z_{l_1 l_2}$ and $z_{l_1 l_2}$ is constrained to be nonnegative, the optimal value of $z_{l_1 l_2}$ is exactly the realized distance associated with the halls assigned to l_1 and l_2. Multiplying by c_{l_1,l_2} in the objective recovers the original walking-cost contribution. Restricting the z variables to pairs with $c_{l_1,l_2} > 0$ is sufficient, since pairs with zero shared enrollment do not contribute to the objective.

Compared with the standard disaggregated linearization that introduces one variable for each tuple (l_1, l_2, h_1, h_2), formulation (8)–(11) is substantially more compact in the number of additional variables. In addition, placing the shared-enrollment weight c_{l_1,l_2} in the objective rather than in the linking constraints yields a better-scaled coefficient matrix while preserving the exact same optimal assignment. However, this compactness does not necessarily imply a stronger linear-programming relaxation.

One way to tighten Constraint (9) is to exploit the fact that each lecture is assigned to exactly one hall and add

$$z_{l_1 l_2} \geq d_{h_1,h_2}\big(x_{l_1 h_1} - 1 + \sum_{h: d_{h_1,h} \geq d_{h_1,h_2}} x_{l_2 h}\big), \qquad \forall (l_1,l_2) \in A', h_1 \in H(l_1), h_2 \in H(l_2). \tag{12}$$

This is a valid inequality. Fix (l_1, l_2), h_1, and h_2, and consider any integer-feasible solution. If $x_{l_1 h_1} = 0$, then the term in parentheses is at most zero, because $\sum_h x_{l_2 h} = 1$ and therefore $\sum_{h: d_{h_1,h} \geq d_{h_1,h_2}} x_{l_2 h} \leq 1$. In that case (12) is implied by the nonnegativity of $z_{l_1 l_2}$. Now suppose that $x_{l_1 h_1} = 1$, and let g be the unique hall assigned to l_2. Then the summation term is equal to 1 if $d_{h_1,g} \geq d_{h_1,h_2}$ and 0 otherwise. Hence the right-hand side of (12) is either 0 or d_{h_1,h_2}. In the

latter case, the realized distance is $d_{h_1,g}$, and since $d_{h_1,g} \geq d_{h_1,h_2}$ by construction, we obtain

$$d_{h_1,g} \geq d_{h_1,h_2}.$$

Therefore (12) cannot cut off any integer-feasible assignment, although it may cut fractional solutions and thereby strengthen the linear relaxation.

By symmetry, one may also add

$$z_{l_1 l_2} \geq d_{h_1,h_2}\Big(\sum_{h:d_{h,h_2} \geq d_{h_1,h_2}} x_{l_1 h} - 1 + x_{l_2 h_2}\Big), \qquad \forall (l_1,l_2) \in A', h_1 \in H(l_1), h_2 \in H(l_2). \tag{13}$$

The same validity argument applies after interchanging the roles of l_1 and l_2. Hence (12) and (13) may be imposed together, potentially yielding a stronger linear relaxation.

The original strong cut can also be enlarged on the l_1 side. For fixed (l_1, l_2), h_1, and h_2, let

$$S_2(h_1,h_2) = \{\, h \in H(l_2) : d_{h_1,h} \geq d_{h_1,h_2} \,\}$$

and define

$$S_1^+(h_1,h_2) = \{\, g \in H(l_1) : d_{g,h} \geq d_{h_1,h_2} \text{ for all } h \in S_2(h_1,h_2) \,\}.$$

Then

$$z_{l_1 l_2} \geq d_{h_1,h_2}\left(\sum_{g \in S_1^+(h_1,h_2)} x_{l_1 g} - 1 + \sum_{h \in S_2(h_1,h_2)} x_{l_2 h}\right), \qquad \forall (l_1,l_2) \in A',\ h_1 \in H(l_1),\ h_2 \in H(l_2). \tag{14}$$

This inequality is valid because whenever the assigned hall of l_1 lies in $S_1^+(h_1,h_2)$ and the assigned hall of l_2 lies in $S_2(h_1,h_2)$, every cross-pair between the two sets is at distance at least d_{h_1,h_2}. Otherwise the right-hand side of (14) is nonpositive. Since $h_1 \in S_1^+(h_1,h_2)$, inequality (14) dominates (12) pointwise.

The strengthening can in fact be strict on a large family of fractional points. Fix $(l_1,l_2) \in A'$ with $c_{l_1,l_2} > 0$, and let $A = \{a_1,\ldots,a_p\} \subseteq H(l_1)$ and $B = \{b_1,\ldots,b_q\} \subseteq H(l_2)$ with $p,q \geq 2$. Assume that $d_{a_i,b_j} = D > 0$ for every $a_i \in A$ and $b_j \in B$. Consider the fractional point defined by

$$x_{l_1,a_i} = \frac{1}{p} \quad \forall a_i \in A, \qquad x_{l_2,b_j} = \frac{1}{q} \quad \forall b_j \in B,$$

all other assignment variables for l_1 and l_2 equal to zero, and $z_{l_1 l_2} = 0$. Then, for every $a_i \in A$ and $b_j \in B$,

$$x_{l_1,a_i} + x_{l_2,b_j} - 1 = \frac{1}{p} + \frac{1}{q} - 1 \leq 0,$$

so all inequalities (9) are satisfied. However, for any fixed $a_i \in A$ and $b_j \in B$, one has

$$\sum_{h:d_{a_i,h} \geq d_{a_i,b_j}} x_{l_2 h} \geq \sum_{h \in B} x_{l_2 h} = 1,$$

and therefore (12) implies

$$z_{l_1 l_2} \geq D\left(\frac{1}{p} - 1 + 1\right) = \frac{D}{p} > 0,$$

which contradicts $z_{l_1 l_2} = 0$. Hence (12) cuts off an infinite family of fractional solutions that remain feasible under (9).

3.3 Constraint Programming Formulation

A constraint-programming (CP) formulation can be made more compact by replacing the binary assignment variables with one integer decision variable a_l for each lecture $l \in L$, where a_l denotes the hall assigned to lecture l and its domain is restricted to the compatible set $H(l)$. The model can then be written as

$$\min \quad \sum_{(l_1,l_2)\in A'} c_{l_1,l_2}\, d_{a_{l_1},a_{l_2}} \tag{15}$$

$$\text{s.t.} \quad a_l \in H(l), \qquad \forall l \in L, \tag{16}$$

$$\text{AllDifferent}(\{a_l : l \in L(t)\}), \qquad \forall t \in T. \tag{17}$$

Equation (15) is the CP objective. It minimizes the total walking burden by summing, over all relevant successor pairs, the shared enrollment c_{l_1,l_2} multiplied by the distance between the halls selected by a_{l_1} and a_{l_2}.

Constraint (16) defines the assignment variables. Each lecture is assigned directly to one compatible hall, so the capacity requirement is enforced through the domain of a_l rather than through separate binary variables and assignment equalities.

Constraint (17) enforces temporal feasibility. For every time unit t, all lectures active during that time must take distinct hall values, which ensures that no hall is used simultaneously by two overlapping lectures.

This formulation is a compact high-level CP model. In an implementation based on a CP-SAT solver such as OR-Tools, the hall assignments would still be represented directly by integer variables, and mutual exclusivity could be enforced by `AddAllDifferent`. The objective term $d_{a_{l_1},a_{l_2}}$ may be viewed as concise mathematical shorthand for indexing the distance matrix by decision variables.

If one wants a lower-level encoding that is closer to an actual CP-SAT implementation, one may introduce for each successor pair $(l_1, l_2) \in A$ an auxiliary integer variable u_{l_1,l_2} representing the realized walking distance, together with constraints enforcing

$$u_{l_1,l_2} = d_{a_{l_1},a_{l_2}}. \tag{18}$$

Then the objective in (15) can be replaced by

$$\min \sum_{(l_1,l_2)\in A'} c_{l_1,l_2}\, u_{l_1,l_2}, \tag{19}$$

and the relation (18) can be encoded in CP-SAT by an element constraint or an allowed-assignment table.

Extensions with Additional Room Criteria. The base model isolates the walking-distance component of the lecture-hall assignment problem. In practical applications, one may wish to account for additional room-quality considerations without changing the main combinatorial structure.

These extensions also clarify the relation between the present model and the room-allocation approach of Antunes-Batista et al. (2026). Once feature requirements are incorporated into $H(l)$ and additional assignment penalties are allowed, the model moves closer to practical room-allocation formulations of the type studied there. However, the present framework remains distinct: student movement enters through a quadratic shared-enrollment walking term, capacity and overlap remain hard feasibility constraints, and the additional room-quality effects enter as linear assignment penalties within a single exact optimization model rather than as separate objectives in a multiobjective evolutionary framework.

First, hard room-feature requirements can be incorporated directly through the compatibility sets. Let F_l^{req} denote the set of mandatory features required by lecture l (for example, a laboratory session that must be assigned to a lab room), and let F_h denote the feature set of hall h. Then one may redefine

$$H(l) = \{h \in H : u_h \geq s_l,\ F_l^{\text{req}} \subseteq F_h\}.$$

No further change in the formulations is needed: the nonlinear model, the MILP reformulation, and the CP model already enforce assignment only to halls in $H(l)$.

Second, soft penalties can be added for room-feature mismatch and wasted capacity. Since both effects depend only on the assignment of lecture l to hall h, they may be combined into a single coefficient $p_{lh} \geq 0$, representing the total linear penalty associated with that assignment. For example, p_{lh} may aggregate penalties for missing desirable room features and for excess unused capacity under a chosen measure. The nonlinear objective (2) can then be extended to

$$\min \sum_{(l_1,l_2)\in A'} c_{l_1,l_2} \sum_{h_1\in H(l_1)} \sum_{h_2\in H(l_2)} d_{h_1,h_2}\, x_{l_1h_1} x_{l_2h_2} + \sum_{l\in L} \sum_{h\in H(l)} p_{lh} x_{lh}. \tag{20}$$

These additional terms are linear in the assignment variables, so they preserve the quadratic-plus-linear structure of the model. Consequently, the compact MILP reformulation is obtained simply by adding the same linear terms to (8), while keeping all linking constraints and valid inequalities unchanged. Likewise, in the CP model one may add per-lecture assignment costs indexed by the chosen hall. Thus these extensions move the model closer to practical room-allocation settings while preserving the exact-solution framework developed in this paper.

4 Numerical experiments

This section will evaluate the computational behavior of the proposed formulations and algorithms and quantify the contribution of each modeling enhancement. The numerical study should be organized to separate formulation strength, preprocessing effects, and solver behavior across instance classes.

4.1 Experimental goals

The numerical study should address four main questions: (i) how much stronger the compact MIP reformulation is than the baseline nonlinear model, (ii) how much the valid inequalities improve

the linear relaxation and the branch-and-bound performance, (iii) how effective the preprocessing procedures are in shrinking the compatibility sets and accelerating solution, and (iv) how the MIP and CP approaches compare on instances of increasing size and density.

4.2 Test instances

The experiments should include a synthetic benchmark family generated from controlled parameter ranges. A natural design is to vary the number of halls, the lecture density, and the seed. For example, one may consider several hall sizes (such as 10, 20, 30, 40, and 50 halls), several density levels, and a fixed set of random seeds for each parameter combination. If institutional data become available, a final subsection may report results on one or more realistic instances using the same methodology.

4.3 Compared formulations and algorithmic variants

The comparison should include at least the following variants:

1. the baseline nonlinear formulation;
2. the compact MIP reformulation without strengthening;
3. the compact MIP reformulation with valid inequalities;
4. the compact MIP reformulation with valid inequalities and preprocessing;
5. the CP formulation;
6. optionally, the extended model with additional linear assignment penalties.

If both lightweight and full preprocessing are retained, they should be tested as separate variants.

4.4 Performance measures

For each run, the numerical study should report the objective value, runtime, optimality gap at the time limit, and a solved-to-optimality indicator. For MIP-based variants, it is also useful to report the root relaxation bound, the number of branch-and-bound nodes, and the reduction in compatibility-set size produced by preprocessing. Aggregate statistics should include means, medians, and success rates over the seed set.

4.5 Experiment blocks

The numerical section can be organized into the following blocks. The first block should compare the baseline nonlinear model against the compact MIP reformulation to isolate the value of linearization.

The second block should quantify the marginal benefit of the strengthening inequalities. The third block should evaluate lightweight versus full preprocessing. The fourth block should compare the best MIP variant with the CP formulation. A final block may analyze scalability by plotting runtime and solution quality against instance size and density.

4.6 Implementation and reproducibility

This subsection should document the hardware, solver versions, parameter settings, and time limits used in the experiments. It should also state the random-seed policy and the exact generator settings so that the experiments can be reproduced. If the experiments rely on the current code base, the subsection may describe the implemented generator, the preprocessing options, and the solver configurations used for each formulation.

References

Sina Abdipoor, Razali Yaakob, Say Leng Goh, and Salwani Abdullah. Meta-heuristic approaches for the university course timetabling problem. *Intelligent Systems with Applications*, 19:200253, 2023. doi: 10.1016/j.iswa.2023.200253.

Bruno Antunes-Batista, Soumen Atta, Vítor Basto-Fernandes, Iryna Yevseyeva, and Michael T. M. Emmerich. Multiobjective optimization approaches for room allocation in university course timetabling. *Procedia Computer Science*, 278:663–670, 2026. doi: 10.1016/j.procs.2026.03.037.

Rainer E. Burkard, Eranda Çela, Panos M. Pardalos, and Leonidas S. Pitsoulis. The quadratic assignment problem. In *Handbook of Combinatorial Optimization*, pages 241–338. Kluwer Academic Publishers, 1998.

Edmund K. Burke and Sanja Petrovic. Recent research directions in automated timetabling. *European Journal of Operational Research*, 140(2):266–280, 2002. doi: 10.1016/S0377-2217(02)00069-3.

Michael W. Carter and Craig A. Tovey. When is the classroom assignment problem hard? *Operations Research*, 40(1-Supplement-1):S28–S39, 1992. doi: 10.1287/opre.40.1.S28.

Matthew Davison, Ahmed Kheiri, and Konstantinos G. Zografos. Modelling and solving the university course timetabling problem with hybrid teaching considerations. *Journal of Scheduling*, 28(2):195–215, 2025. doi: 10.1007/s10951-024-00817-w.

Tjalling C. Koopmans and Martin Beckmann. Assignment problems and the location of economic activities. *Econometrica*, 25(1):53–76, 1957. doi: 10.2307/1907742.

Eliane M. Loiola, Nair Maria Maia de Abreu, Paulo Oswaldo Boaventura-Netto, Peter Hahn, and Tania Querido. A survey for the quadratic assignment problem. *European Journal of Operational Research*, 176(2):657–690, 2007. doi: 10.1016/j.ejor.2005.09.032.

Efstratios Rappos, Eric Thi'emard, Stephan Robert, and Jean-François Hêche. A mixed-integer programming approach for solving university course timetabling problems. *Journal of Scheduling*, 25(4):391–404, 2022. doi: 10.1007/s10951-021-00715-5.

Sartaj Sahni and Teofilo Gonzalez. P-complete approximation problems. *Journal of the ACM (JACM)*, 23(3):555–565, 1976.

Andrea Schaerf. A survey of automated timetabling. *Artificial Intelligence Review*, 13(2):87–127, 1999. doi: 10.1023/A:1006576209967.

Camilo Torres-Ovalle, Jairo R. Montoya-Torres, Carlos Quintero-Araujo, and Angelica Sarmiento. University course scheduling and classroom assignment. *Ingeniería y Universidad*, 18(1):59–75, 2014. doi: 10.11144/Javeriana.IYU18-1.ucsc.